Knowledge Graph-Assisted LLM Post-Training for Enhanced Legal Reasoning

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Abstract

LLM post-training has primarily relied on large text corpora and human feedback, without capturing the structure of domain knowledge. This has caused models to struggle dealing with complex reasoning tasks, especially for high-stakes professional domains. In Law, reasoning requires deep understanding of the relations between various legal concepts, a key component missing in current LLM post-training. In this paper, we propose a knowledge graph (KG)-assisted approach for enhancing LLMs' reasoning capability in Legal that is generalizable to other high-stakes domains. We model key legal concepts by following the **IRAC** (Issue, Rule, Analysis and Conclusion) framework, and construct a KG with 12K legal cases. We then produce training data using our IRAC KG, and conduct both Supervised Fine-Tuning (SFT) and Direct Preference Optimization (DPO) with three state-of-the-art (SOTA) LLMs (30B, 49B and 70B), varying architecture and base model family. Our post-trained models obtained better average performance on 4/5 diverse legal benchmarks (14 tasks) than baselines. In particular, our 70B DPO model achieved the best score on 4/6 reasoning tasks, among baselines and a 141B SOTA legal LLM, demonstrating the effectiveness of our KG for enhancing LLMs' legal reasoning capability.

1 Introduction

Large language models (LLMs), such as GPT OpenAI [2023], Claude Anthropic [2023] and Llama Touvron et al. [2023], have shown remarkable performance on a wide range of natural language tasks. However, when the target applications are in highly specialized domains, such as Medicine, Finance, Law, etc., general-purpose LLMs often struggle with specialized terminologies and domain-specific reasoning needs. Consequently, researchers have begun to build domain-specific variants by specifically pre-training and fine-tuning on corpora from the corresponding target fields Singhal et al. [2023, 2025], Xie et al. [2024], Colombo et al. [2024].

While training and continuous fine-tuning on raw text capture the generation of domain language, they are insufficient to encode the structure of domain knowledge, especially on domain-specific reasoning tasks. Recent reasoning models work best for areas, such as mathematics and programming, where verifiable rewards enable a straightforward approach to reinforcement learning; however, areas like Medicine or Law require alternative forms of reasoning, such as argumentative reasoning or inference in the presence of high and inherent uncertainty, by using concepts of the corresponding domains, such as symptoms, events, and legal rules.

More importantly, these areas require deep understanding of the relations between abstract concepts. Financial market-event analysis adopts a cause-effect-impact-conclusion chain; clinical reasoning follows a symptom-diagnosis-treatment-outcome flow; in Legal, judicial reasoning employs the fact-rule-conclusion process. Encoding these reasoning patterns in a structured form offers a principled way to augment LLMs' inductive bias and to reduce hallucinations or misattributions.

In this paper, we focus on the legal domain where even when models are trained on vast corpora of case law, statutes, regulations, and secondary literature, they may produce misattributions, hallucinations, and logically incoherent arguments Magesh et al. [2025], and specific tasks are often better served by smaller specialized models Chalkidis [2023], Jayakumar et al. [2023], Shu et al. [2024]. The root of this problem lies not in the quantity of text for LLM training but in the structure of legal knowledge: legal prose is dense, highly contextual, and governed by rules that change over time, calling for a clearer textual representation that makes the element of the reasoning process more accessible.

In order to address this gap, we start with the formal modeling of legal concepts, which then leads to generating textual training data for LLM post-training. Our approach explicitly encodes the reasoning patterns found in judicial opinions in order to facilitate LLM training. The cornerstone of legal reasoning is the **IRAC** framework, ***I****ssue*, ***R****ule*, ***A****nalysis* and ***C****onclusion*, which is taught in law schools and encapsulates the typical flow of legal argumentation Brand and White [1976]. By constructing an **IRAC Knowledge Graph** (IRAC KG) that captures the relationships between issues, governing rules, factual applications, and conclusions, we aim to provide a structured substrate from which an LLM can learn to reason more faithfully.

Our work equips LLMs with a structured substrate for legal reasoning, but with two distinct features. First, instead of focusing on creating legal KGs with specific entities, such as Person, Organization, etc., our goal is to model the key legal concepts by following the IRAC framework. Second, rather than treating the graph merely as retrieval scaffolding, we post-train LLMs directly on graph-derived supervision signals, and show that this yields measurable gains on diverse legal tasks. Our contributions are threefold:

1. **IRAC KG construction.** We extract IRAC components from a large corpus of 12K legal case opinions, and encode them as nodes and typed edges in a graph, preserving the causal dependencies that underlie legal reasoning.

2. **IRAC KG-based LLM post-training.** By utilizing the KG as a knowledge source, we post-train mid to large LLMs (Qwen3-30B-A3B-Instruct-2507, Llama-3_3-Nemotron-Super-49B-v1_5 and Llama-3.1-70B-Instruct) via both Supervised Fine-Tuning (SFT) and Direct Preference Optimization (DPO) Rafailov et al. [2023] in order to improve their ability to generate coherent legal arguments.

3. **Extensive evaluation.** We assess the trained models on five legal benchmarks: LexGLUE Chalkidis [2023], LegalBench Guha et al. [2023], COLIEE Rabelo et al. [2022], SuperGPQA (Law) Team et al. [2025] and δ-Stance Gupta et al. [2025], demonstrating measurable gains over baseline and SOTA models.

2 Related Work

2.1 LLM-based KG Construction

KG creation pipelines typically consist of two steps: Named Entity Recognition (NER) and Relation Extraction (RE). Prompting-based strategies have proven to be effective, addressing both tasks jointly Edge et al. [2024], Guo et al. [2024] or separately Jimenez Gutierrez et al. [2024], Gutiérrez et al. [2025]. Compared to traditional methods, LLMs are also capable of extracting additional metadata, such as verbose entity descriptions or numerical scores measuring relation strength Guo et al. [2024].

Focusing on KG quality, Zhang and Soh [2024] defined a three-stage solution to ground LLMs into existing KGs, or to canonicalize one derived via open extraction. Han et al. [2024b] trained small models for output quality verification and iteratively improving the quality of the original KG, while Zhu et al. [2024] developed a multi-agent framework to guide this refinement process by integrating external knowledge.

2.2 Graph-based LLM Reasoning

Researchers have utilized LLMs and graphs to address complex reasoning scenarios Pan et al. [2024]. GraphRAG Cui et al. [2023] organizes text corpora as a KG, and reasons over community descriptions to answer queries. This enables LLMs to handle questions that require a general understanding of the corpus and inspired follow-up methods to explore structured representations to improve Retrieval Augmented Generation (RAG) pipelines Peng et al. [2024], Han et al. [2024a].

A recurring theme in several of these works is to present the retrieved information to LLMs in structured formats, e.g., triples or relational paths, which has improved reasoning fidelity and quality Ma et al. [2025], Sun et al. [2024], Liu et al. [2024], Wen et al. [2024]. Rather than using graphs for retrieving structured information, Cheng et al. [2024] reformulate the input query as a graph, and propose a multi-stage prompting pipeline to reason over the input graph for query answering. The authors claimed that the method reflects how humans often approach complex problems.

In addition to fostering reasoning, graphs were also used to model the reasoning process itself. Both Tree-of-Thoughts Yao et al. [2023] and Graph-of-Thoughts Besta et al. [2024] improved over the classical Chain-of-Thought idea that lacks the capability of expressing complex thought processes, such as backtracking on unpromising steps or combining ideas from different branches.

2.3 KG-LLM Synergy in the Legal Domain

Tang et al. [2024c] represented legal cases and charges as nodes, and connected them using existing metadata, such as case-case citations and case-charge relationships. Dhani et al. [2021] developed a more complex pipeline to extract entities and relations from Indian cases, judgments and laws using traditional NER and RE techniques for the purpose of finding similar cases.

More recently, Chen et al. [2024] used an LLM to extract triples from Chinese cases and criminal law articles by following a pre-defined schema. Similar to our approach, Tang et al. [2024b,a] relied on an LLM to represent case legal issues and facts as a KG (though without considering precedents). This offered an initial, though partial, representation of a case via the IRAC framework.

While being effective in solving important tasks, Hannah et al. [2025] showed that serious legal implications might arise from LLM responses, and proposed a KG-RAG solution to address potential legal concerns. Similarly, Colombo [2024] combined KG and RAG to develop an Italian legislative platform to support law analysis, drafting, landscape identification and complexity monitoring. Finally, ChatLaw Cui et al. [2023] is a multi-agent system for legal consulting, and utilizes KGs in intermediate steps to assist answering user requests.

Instead of extracting specific entities (e.g., Person, Location, etc.), our work builds KGs that model key legal concepts (such as Facts and Rules). Also, different from existing methods that combine LLMs with graphs, our novelty lies in the use of graph structure for domain-specific LLM post-training. Finally, while previous approaches studied partial modeling of legal reasoning, our IRAC KG offers a more comprehensive coverage of the important legal reasoning components.

3 System Overview and KG Design

3.1 Overview

Figure 1 demonstrates an overview of our entire knowledge graph-assisted LLM post-training system. The first step is to extract the key concepts from legal cases (from U.S. jurisdictions) by utilizing an LLM and a prompt (Appendix E.1) that includes the KG schema (Section 3.2). Then, we produce training data for both Supervised Fine-Tuning (SFT) and Preference Training. During data generation, in addition to prompts, we also employed an LLM for explanation generation (Section 4.1) and as an LLM Judge (Section 4.2).

3.2 Knowledge Graph Design

Different from prior works that extract specific entity names (e.g., Person, Company, Location, etc.), our KG focuses on extracting the key legal concepts (such as Facts and Legal Issues) in order to facilitate legal reasoning. Figure 2 shows the design of our legal KG, a key component of our proposed system, by following the IRAC framework.

IRAC is commonly adopted for conducting legal analysis Burton [2017]. Even though there are variations, such as ILAC (Issue, Law, Analysis, Conclusion), IRAACP (Issue, Rule, Apply, Apply, Conclusion, Policy), etc., the core idea of legal reasoning centers around the same main concepts, i.e., Issue, Rule, Analysis and Conclusion. We consulted four legal experts on the design in order to capture the core of the legal reasoning process. Our IRAC KG includes the following components:

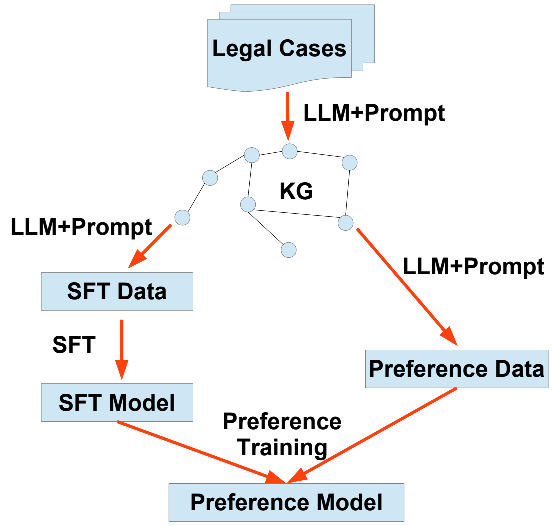

Figure 1 System overview.

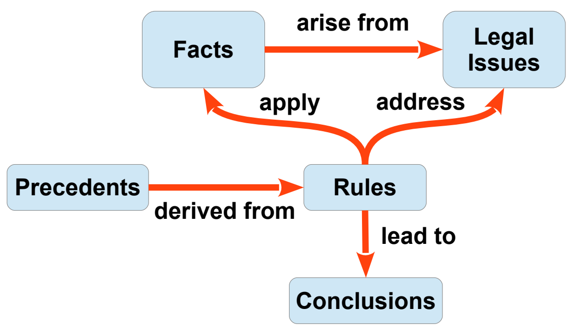

Figure 2 Schema of our IRAC legal knowledge graph.

- Legal Issues (**I**): This represents legal issues that the different parties (e.g., plaintiffs, defendants, judges, attorneys, etc.) are addressing in a legal case. Facts are things that actually happened, and the reason we have a legal case is due to the facts. Therefore, we have *Legal Issues* **arise from** *Facts* in our schema.
- Rules (**R**): This refers to legal rules. In the legal domain, prior cases (also known as precedents) that are sufficiently representative become rules that courts will reference to decide on future cases. Therefore, *Rules* are **derived from** *Precedents*.
- Analysis or Application (**A**): This denotes the legal reasoning process. When deciding on a legal case, courts will **apply** *Rules* to *Facts* in order to **address** the *Legal Issues*.
- Conclusions (**C**): The analysis that courts carry out will eventually **lead to** certain *Conclusions*. These are courts' rulings on the case.

4 Training Data Generation

Our goal is to study whether such legal reasoning KGs would benefit LLMs on diverse tasks by conducting both SFT and Preference Optimization. In this section, we present how we utilized our IRAC KG to produce the corresponding training data.

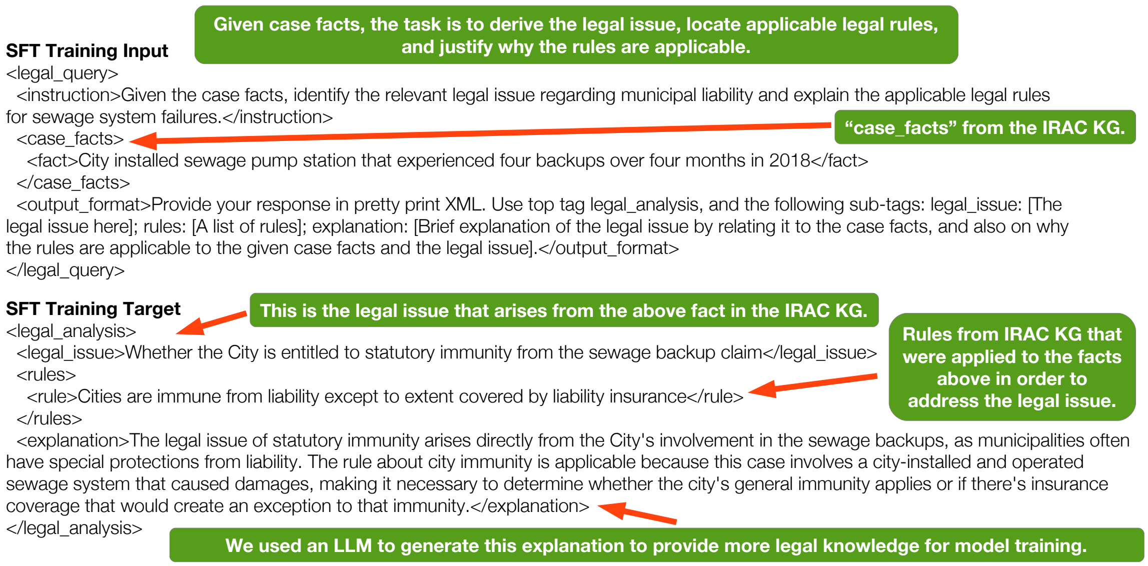

Figure 3 An example of the Rules SFT dataset.

4.1 The Rules SFT Dataset

Figure 3 illustrates a concrete example of our SFT dataset. The core idea is to simulate how courts identify rule(s) that are applicable to addressing a specific legal issue. In SFT input, the main component is *case_facts*, which is the foundation to have a legal case. Then, in the training target, we teach a model how to derive the legal issue that arises from the fact(s), and further identify the applicable rule(s). In addition, we also provide an explanation on why the rule(s) are applicable to the legal issue.

Algorithm 1 depicts this SFT data generation process. At Line 2, $Facts$ consists of all facts related to

Algorithm 1 GenSFT(KG, l_i, LLM), KG represents the IRAC KG; l_i denotes a legal issue in KG; a large language model LLM produces an explanation on the applicability of certain rule(s) to l_i.

1: $Rules \leftarrow \emptyset$
2: $Facts \leftarrow GetRelatedFacts(KG, l_i)$
3: **for all** $f \in Facts$ **do**
4: $\quad R_f \leftarrow GetRsViaApply(KG, f)$
5: $\quad Rules \leftarrow Rules \cup R_f$
6: **end for**
7: $OtherRs \leftarrow GetRsViaAddress(KG, l_i)$
8: $Rules \leftarrow Rules \cup OtherRs$
9: $Exp \leftarrow Explain(LLM, l_i, Rules, Facts)$
10: **return** $Facts, Rules, Exp$

legal issue l_i, via the ***arise from*** relation in Figure 2. Next, from Line 3 to 8, we find the set of applicable rules to l_i both directly and indirectly. On one hand, at Line 4, for each fact f related to l_i, Function $GetRsViaApply(KG, f)$ finds the rules that were applied to f by following the path: *Rules*$\rightarrow$ ***apply*** $\rightarrow$ *Facts*. On the other hand, at Line 7, by following the *Rules* $\rightarrow$ ***address*** $\rightarrow$ *Legal Issues* chain in Figure 2, Function $GetRsViaAddress(KG, l_i)$ gathers additional rules identified by the court that could be used to address l_i but may not actually be applied to any facts, e.g., rules from similar precedents. As demonstrated in Figure 3, the retrieved facts become SFT input, while the rules and the legal issue itself are part of the training target.

In addition, at Line 9, we prompt (Appendix E.2) an LLM to produce a concise explanation on why the rules are applicable to the legal issue l_i, hoping to provide more legal knowledge to downstream training tasks; this explanation constitutes another section of the training target. We conducted a quality review of 15

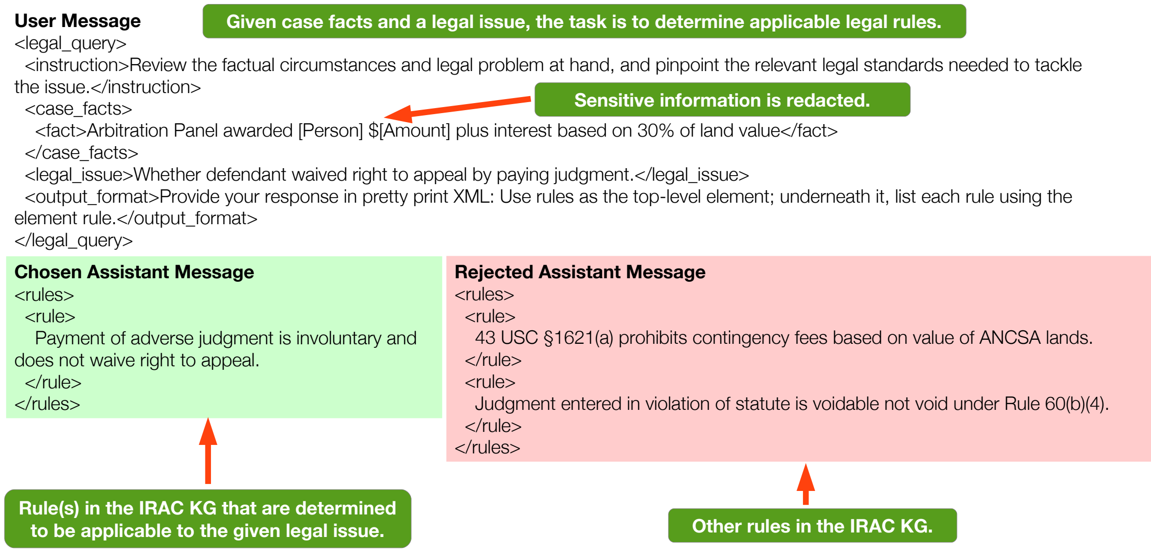

Figure 4 An example of the Rules preference dataset.

SFT examples with three subject matter experts (SMEs), where each example was labeled by a single SME. Among these, 11, 4 and 0 examples were graded as *Correct*, *Correct with minor issues* and *Wrong* respectively, verifying the quality of our SFT dataset.

Entity	**Claude Sonnet 3.5 KG**				**Llama-3.1-405B-Instruct KG**			
	Good	**Acceptable**	**Poor**	**M**	**Good**	**Acceptable**	**Poor**	**M**
Fact	68%	5%	2%	26%	50%	9%	6%	35%
Legal Issue	63%	0%	0%	37%	75%	5%	10%	10%
Rule	69%	0%	4%	27%	29%	0%	33%	38%
Conclusion	68%	0%	0%	32%	60%	0%	10%	30%
Relation	**Pass**		**Fail**		**Pass**		**Fail**	
	92%		8%		59%		41%	

Table 1 IRAC KG quality review. *Fact*, *Legal Issue*, *Rule* and *Conclusion* correspond to the entity types in our IRAC KG in Figure 2; we report a single number for all relations. For entities, our SMEs labeled them as ***Good*** (high-quality with potentially minor inaccuracies), ***Acceptable*** (with noticeable issues but still useful) and ***Poor*** (with significant errors or completely incorrect); our SMEs also checked whether the KG missed any entities (***M***) by reading the entire case. As for relations, our review only examines whether they are correct or not. Note that if any of the entities involved in a relation is judged as ***Poor***, then this relation will be labeled as ***Fail***. We determined these quality review criteria together with our domain experts.

4.2 The Rules Preference Dataset

In addition to SFT, we also performed Preference Optimization in this work. Similar to the SFT dataset, we produced our preference data by utilizing the IRAC KG with a focus on teaching the models to correctly identify the proper legal rules to cope with a given legal issue. Figure 4 shows a concrete example of this dataset. Different from the above SFT data, our preference training input (i.e., *User Message*) includes both *case_facts* and *legal_issue*, with chosen and rejected legal rules as the training target. Our hypothesis is that by teaching models how to select the appropriate rules for different legal matters, they would be able to gain more knowledge on how courts reason in a variety of legal situations.

We formally present the data generation process in Algorithm 2. For preference training, we need to provide both the chosen and the rejected rules. Similar to SFT data generation, at Line 1, we first obtain facts ($Facts$) related to the legal issue l_i. Then, from Line 2 to 8, we find the chosen rules ($Chosen\_Rs$) that are either directly (via the ***address*** relation) or indirectly (via the combination of the ***apply*** and the ***arise from***

Algorithm 2 GenPref(KG, l_i, LJ), KG denotes the IRAC KG; l_i is a legal issue in KG; an LLM Judge LJ determines a rule's applicability to l_i.

1: $Facts \leftarrow GetRelatedFacts(KG, l_i)$
2: $Chosen\_Rs \leftarrow \emptyset$
3: **for all** $f \in Facts$ **do**
4: $\quad R_f \leftarrow GetRsViaApply(KG, f)$
5: $\quad Chosen\_Rs \leftarrow Chosen\_Rs \cup R_f$
6: **end for**
7: $OtherRs \leftarrow GetRsViaAddress(KG, l_i)$
8: $Chosen\_Rs \leftarrow Chosen\_Rs \cup OtherRs$
9: $Rejected\_Rs \leftarrow All\_Rs \setminus Chosen\_Rs$
10: **for all** $r \in Rejected\_Rs$ **do**
11: $\quad$ **if** $Applicable(LJ, Facts, l_i, r)$ **then**
12: $\qquad Rejected\_Rs \leftarrow Rejected\_Rs \setminus \{r\}$
13: $\quad$ **end if**
14: **end for**
15: **return** $Facts, Chosen\_Rs, Rejected\_Rs$

relations) connected to l_i. In order to produce the rejected rules, from Line 9 to 14, we use an LLM judge (Appendix E.3) to determine whether each of the non-chosen rules is truly not applicable. The intuition is that even though a rule is not directly or indirectly connected to l_i in the KG, it may be due to an imperfect KG generation process. The output of the LLM Judge could be: *Yes*/*No*/*Potentially*, and we only keep the rules that are determined as non-applicable (i.e., the LLM Judge outputs *No*).

4.3 Model Training

In our work, we conduct both SFT and Preference Optimization, DPO Rafailov et al. [2023] specifically. By giving models the case facts, SFT (Figure 3) teaches the models how to identify the legal issue, locate the applicable rules, and learn the rationale behind these materials (*Exp* at Line 9 in Algorithm 1). Different from SFT, our preference data (Figure 4) focuses on teaching the models how to conduct legal reasoning by contrasting the applicable and non-applicable legal rules with respect to a variety of legal matters. For DPO, we adopted the default Loss function with a small NLL term on the chosen answers Pang et al. [2024].

We emphasize that the above training data is not directly related to our evaluation benchmarks (Section 5.2). Our primary goal is to provide the models with high-quality domain knowledge via different training paradigms in order to study whether the trained models would be able to gain improved capability on diverse tasks in the same domain.

5 Evaluation

5.1 IRAC KG Generation and Quality Review

We randomly sampled ∼12K cases, with 220 cases from each U.S. jurisdiction, and divided them into Train and Validation for model training purposes with a ratio of 10:1. We produced KGs with both Claude Sonnet 3.5 and Llama-3.1-405B-Instruct. Two SMEs then reviewed the Sonnet KG of 18 cases and the Llama KG of 8 cases[1], with each entity and relation examined by a single SME.

Table 1 shows that the Sonnet KG has superior quality on most entity types (except for *Legal Issue*), by taking into account both *Good* and *Acceptable*. Note that the Llama KG generally has notably higher ratios of entities labeled as *Poor*. In terms of relations, the Sonnet KG also showed much higher quality, with only 8% of relations labeled as *Fail* compared to that of 41% of the Llama KG.

One drawback of both KGs is that they miss a decent percentage of entities. In a legal case, not all facts or

[1]Due to its inferior quality, we stopped the review process.

issues are considered critical for courts to decide their rulings. A legal issue may be mentioned in passing without further discussions; also, some facts may merely be included in a case while not being essential to the core issues of the case. In such situations, missing entities have little impact in understanding the case. We leave it to future work to improve our KG's comprehensiveness.

Given the above quality review, we adopted the IRAC KG from Claude Sonnet 3.5 for producing the post-training datasets. We also utilized Sonnet 3.5 to generate the explanation (Exp at Line 9 in Algorithm 1) for the SFT dataset, and employed it as the LLM Judge (LJ at Line 11 in Algorithm 2) during preference data generation.

5.2 Evaluation Benchmarks

In order to verify the effectiveness of our proposed IRAC KG, we carried out extensive experiments on diverse legal datasets (detailed in Appendix A).

LexGLUE[2] Chalkidis [2023] is a well-adopted legal benchmark, covering European and U.S. jurisdictions. It mainly consists of (multi-class and multi-label) classification tasks, together with a multiple choice question answering task; each task includes 1,000 samples. The number of classes range from 10 to 100, presenting substantial challenges to legal AI systems.

LegalBench Guha et al. [2023] is a recent benchmark that mainly focuses on multiple choice question answering. In addition, it also includes several information extraction tasks, such as extracting plaintiffs and defendants from case excerpts. It includes a total of 162 tasks, and we removed *rule_ qa* whose evaluation requires manual review according to the LegalBench team.

COLIEE Rabelo et al. [2022] is a well-known competition for legal systems, with two major types of tasks: retrieval (of relevant cases or statues) and entailment. For retrieval, it is to find supporting prior cases to help decide on a new case in Canada (Task-1) or to locate relevant law articles (statutes) for Japanese legal queries (Task-3). For entailment, instead of requiring the systems to locate the relevant prior cases or statues, the input already includes this information, and the systems should decide whether the following entailments hold: $<$paragraph (from a prior legal case) $\models$ court decision$>$ (Task-2) or $<$statute $\models$ legal_query$>$ (Task-4). Given that the test set is relatively small for each specific year, we combined the data from all previous years for our evaluation.

SuperGPQA Team et al. [2025] consists of graduate-level multiple choice questions for a broad range of domains, e.g., Medicine, History, etc. Our evaluation focuses on the 656 questions in *Law*, where each question contains 9 $\sim$ 10 choices.

δ-**Stance** Gupta et al. [2025] is a large-scale dataset to specifically evaluate AI systems' legal reasoning capability. In U.S. legal cases, introductory signals (e.g., *see, contra, see generally, etc.*) between two text snippets often indicate certain important relations. This dataset consists of $\sim$14K triples of $<$snippet A, signal, snippet B$>$. We consulted SMEs to group such triples into four higher-level classes (Appendix B): directly supports, indirectly supports, contradicts and background. We also produced a 3-class version where we removed the "background" category that does not provide as much value as the other three according to SMEs.

5.3 Model Training Details

We post-trained (Appendix C) models of different scales: **Qwen3-30B-A3B-Instruct-2507** (Qwen3-30B), **Llama-3_3-Nemotron-Super-49B-v1_5** (Nemotron1.5-49B) and **Llama-3.1-70B-Instruct** (Llama3.1-70B). The first two recent models obtained competitive results on various benchmarks. As for Llama Dubey et al. [2024], the 3.3 model is not as competitive as the 3.1 model (Appendix D), thus we utilized the earlier version.

5.4 Results and Discussion

SFT/DPO vs. OOB. In Table 2, compared to the OOB models, our KG-based post-training achieved improvements on 10, 13 and 12 out of the 14 tasks for Qwen3-30B, Nemotron1.5-49B and Llama3.1-70B

[2] https://github.com/coastalcph/zeroshot_lexglue

Benchmark	Qwen3-30B			Nemotron1.5-49B			Llama3.1-70B			SaulLM
	OOB	SFT	DPO	OOB	SFT	DPO	OOB	SFT	DPO	(141B)
LexGLUE Average	57.9	60.3	**61.6**	57.0	61.2	**62.5**	65.0	66.8	**67.2**	64.4
ECtHR A★	53.4	59.7	62.8	52.8	63.2	66.3	73.6	77.3	77.9	73.6
ECtHR B★	60.2	66.3	69.0	60.9	70.0	73.7	80.0	82.1	82.3	77.6
EUR-LEX★	27.6	28.5	30.5	27.1	29.2	29.5	33.9	32.3	31.8	32.2
LEDGAR★	69.7	69.6	69.6	66.0	66.5	67.2	63.6	66.8	67.5	71.8
SCOTUS★	66.0	65.8	65.9	65.4	66.0	65.4	67.5	68.9	68.8	52.2
CaseHOLD§◊	70.7	71.7	72.0	69.7	72.5	73.2	71.6	73.4	75.1	79.0
COLIEE Average	56.6	59.0	**61.1**	49.4	53.5	**54.2**	57.0	59.8	**60.7**	24.6
Task-1★	39.8	35.0	40.3	20.3	28.3	28.5	30.8	30.6	32.4	0.0
Task-2★◊	52.2	60.0	62.1	48.0	54.6	57.2	57.7	66.0	67.6	25.1
Task-3△	64.3	67.1	67.7	53.6	53.4	53.4	61.3	64.4	64.6	0.0
Task-4§◊	70.2	73.9	74.2	75.9	77.5	77.7	78.4	77.9	78.0	73.1
δ-Stance Average†◊	**45.5**	44.3	43.7	33.9	38.7	**40.9**	48.8	49.7	**50.1**	43.8
3-class	52.7	51.2	50.7	38.8	44.0	46.5	55.1	56.1	56.5	48.8
4-class	38.3	37.3	36.8	29.1	33.4	35.3	42.6	43.2	43.6	38.8
LegalBench▲	74.1	75.3	**75.4**	74.2	75.9	**76.4**	78.0	**79.3**	**79.3**	79.3
SuperGPQA (Law)§◊	42.4	42.1	**42.5**	37.7	37.6	**38.0**	40.3	41.7	**43.8**	28.4

◊: Legal reasoning task.

★: Micro-F1; †: Macro-F1; △: Macro-F2; §: Accuracy; ▲: Balanced Accuracy.

Table 2 Evaluation results (0∼100). OOB denotes the out-of-the-box model without additional training; SFT and DPO represent the models after the corresponding training respectively. For each model, we underscore each task's best performance, and bold the highest average performance; blue background indicates the highest score for a row.

respectively, where the best performance falls on either SFT or DPO. Furthermore, for all three models, DPO training helped obtain the highest average performance on most benchmarks, except for Qwen3-30B on δ-Stance.

DPO vs. SFT. Compared to SFT, DPO did provide additional benefits on 11/14 tasks for Qwen3-30B and Llama3.1-70B, and on 12/14 tasks for Nemotron1.5-49B. Unlike prior works that mostly focus on training smaller models, such as 7B/8B Fan et al. [2025], Wang et al. [2025], Saeidi et al. [2024] and 32B Wen et al. [2025], our results demonstrated the benefits of KG-based DPO training of different and also larger scales: 30B, 49B and 70B. Furthermore, compared to the 30B model, the two larger DPO models exhibited clearer advantages than SFT on 4/6 legal reasoning tasks: CaseHOLD, δ-Stance (both tasks) and SuperGPQA.

SaulLM (141B) Colombo et al. [2024] is a SOTA legal LLM that was trained on a large amount of legal data. Despite post-trained on much smaller SFT (13.5K) and DPO (5.3K) datasets, our models show noticeable benefits. First, while Nemotron1.5-49B OOB is substantially worse than SaulLM on LexGLUE, δ-Stance and LegalBench, its DPO version clearly narrowed the gaps. Furthermore, although Llama3.1-70B OOB already outperforms SaulLM on most benchmarks, our post-training did fully close the gap on LegalBench.

Instruction Following (IF). Since Nemotron1.5-49B is also based on Llama, we only present the scores of Qwen3-30B and Llama3.1-70B in Table 3. In general, both models were able to follow the instructions in our prompts. Furthermore, our post-training did not damage models' IF capability, and in fact, they provided noticeable improvements: Qwen3-30B on LexGLUE, and Llama3.1-70B on COLIEE and SuperGPQA (Law).

We did not employ complex post-processings: We lowercased model outputs, removed punctuations, and performed exact string matching against the groundtruth. Due to our relatively strict metric calculation, compared to other benchmarks, we do observe higher error rates on LexGLUE (mainly classification tasks), where we require the predictions of a test sample to be a subset of the available labels for the corresponding task. For instance, for EUR-LEX (multi-label classification of EU legislative documents), Qwen3-30B OOB predicated *risk management* while only *management* is a valid label. We do not give partial credit in such scenarios.

Benchmark	Qwen3-30B O/S/D	Llama3.1-70B O/S/D
LexGLUE	5.7/3.4/2.1	1.8/2.7/2.2
COLIEE	0.4/0.3/0.2	3.4/1.8/2.1
δ-Stance	0.0	0.0
LegalBench	0.4/0.2/0.2	0.2/0.1/0.2
SuperGPQA	0.0	1.4/0.0/0.2

Table 3 IF error rates (0∼100). We report the average of all tasks for each benchmark. O, S and D represent the OOB, SFT and DPO models respectively; a single number indicates the same rate for all models.

6 Conclusion

In this paper, we proposed a generalizable knowledge graph-assisted approach for enhancing LLMs' legal reasoning capability. We follow the IRAC framework in order to model courts' analysis and reasoning process of legal cases. Furthermore, we conducted both SFT and DPO training with mid to large models (30B, 49B and 70B) by utilizing data generated from the IRAC KG. Our trained models achieved better average performance on 4/5 diverse legal benchmarks than baselines. Compared to SFT, the Llama3.1-70B DPO model exhibited promising results on 3/6 legal reasoning tasks, with 1.7% (CaseHOLD), 1.6% (COLIEE-Task-2) and 2.1% (SuperGPQA Law) higher scores respectively; in particular, it achieved noticeably higher scores than SaulLM, a 141B SOTA legal LLM trained on a large amount of legal data.

Limitations

Our current work is limited to U.S. Law (common law), and the generalizability of our approach to other legal systems warrants further exploration. In addition to legal, the generalizability of our proposed KG-assisted model post-training for other high-stakes domains would also be an important topic. Furthermore, in this work, we produced our data with 12K U.S. cases. It would be interesting to utilize more cases, and also study the impact of additional relations, e.g., citations and overrulings; however, we need to be mindful of the environmental impact, since it would require more GPU hours. Also, given the emergence of other LLMs, we should explore alternatives other than Claude Sonnet 3.5 for KG and data creation, and compare the performance of the resulting models. Finally, although our models showed advantages on diverse legal benchmarks, it would be important to deploy them in front of users to gather feedback on real-world tasks, especially those that require complex legal reasoning. When deploying the trained models for real-world use, it would be ideal to apply certain guardrails to prevent potential malicious usage, especially for the legal domain.

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A Evaluation Benchmarks

Table 4 provides the details of our benchmarks. All our evaluation datasets and the 12K sampled U.S. cases for KG creation (Section 5.1) are in English. All use of these artifacts are consistent with their intended use.

Benchmark	**Task Type**	**Total Tasks**	**Sample Count**	**Public**
LexGLUE [Chalkidis, 2023]	Classification and MCQA	6	1,000 for each task	Yes
LegalBench [Guha et al., 2023]	Yes/No, MCQA and Information Extraction	161	$46 \sim 10$K per task	Yes
COLIEE [Rabelo et al., 2022]	Information Retrieval and Entailment	4	$500 \sim 1.2$K per task	Yes
SuperGPQA [Team et al., 2025]	MCQA	1	656	Yes
δ-Stance Gupta et al. [2025]	Classification	2	14K	Yes

Table 4 Evaluation benchmarks. MCQA stands for Multiple Choice Question Answering. For SuperGPQA, we only evaluated on the *Law* questions.

B Grouping of Introductory Signals

Table 5 shows the grouping of all introductory signals into higher-level classes for our evaluation on the δ-Stance dataset Gupta et al. [2025].

Higher Level Class	Signals
directly supports	"e.g.,", "accord"
indirectly supports	"see", "see also", "cf."
contradicts	"but cf.", "but see" "contra"
background	"see generally"

Table 5 Grouping of introductory signals for δ-Stance. For the 3-class version in our evaluation, we removed the "background" class.

C Model Training and Inference Details

We adopted Transformers Wolf et al. [2020], DeepSpeed[3], Liger Kernel Hsu et al. [2024] and Transformer Reinforcement Learning (TRL)[4] for training. Table 6 shows the detailed training hyperparameters. We did not conduct any parameter search; instead, we followed industry best practices to set the important parameters.

One ml.p4de.24xlarge instance has eight A100 GPUs, and each GPU has 80GB GPU RAM. For Llama-3.1-70B-Instruct, we employed three and four instances for SFT and DPO respectively; for Llama-3_3-Nemotron-Super-49B-v1_5, we utilized two and three instances for SFT and DPO respectively; for Qwen3-30B-A3B-Instruct-2507, we adopted one and two instances for SFT and DPO respectively.

In terms of training time, for Qwen3-30B-A3B-Instruct-2507, Llama-3_3-Nemotron-Super-49B-v1_5 and Llama-3.1-70B-Instruct, it needed 13, 2.5 and 3.5 hours for SFT respectively, and took 6, 2 and 2.5 hours for DPO respectively. For all model training, we trained for five epochs, and used our development set (Section 5.1) for model selection. All our model training is fully reproducible. In total, our model training consumed 320 GPU hours.

We also employed SGLang[5] for inference; due to its non-reproducible nature, we ran each inference twice and reported the average. On our evaluation benchmarks, a single inference run can take up to 70 minutes (the COLIEE benchmark in Appendix A). We ran all inferences on a single instance. All our evaluations consumed a total of 480 GPU hours.

These libraries are open-source, and the base models are available from Hugging Face[6]. The only proprietary artifact we use is Sonnet 3.5. All use of these artifacts are consistent with their intended use.

[3] https://github.com/deepspeedai/DeepSpeed
[4] https://github.com/huggingface/trl
[5] https://github.com/sgl-project/sglang
[6] https://huggingface.co/models

Hyperparameter	SFT	DPO
Learning rate	1e-6	5e-7
Scheduler	linear	
Effective batch size	384	192
Optimizer	Adam	
Attention	FA2	
Liger Kernel	Yes	No
Model precision	bfloat16	
Gradient checkpointing	Yes	
DeepSpeed	Zero-3 without offload	
Hardware	ml.p4de.24xlarge	
Training Data Size	13.5K	5.3K

Table 6 Model training hyperparameters. SFT and DPO represent Supervised Fine-Tuning and Direct Preference Optimization respectively. *Effective batch size* includes gradient accumulation.

D Llama-3.1-70B-Instruct vs. Llama-3.3-70B-Instruct

Table 7 compares the performance between Llama-3.1-70B-Instruct and Llama-3.3-70B-Instruct. First of all, on all benchmarks, Llama-3.1 had higher average performance than Llama-3.3, suggesting that it could be a stronger base model for training. Furthermore, Llama-3.1 also demonstrates better performance on 10 out of the 14 individual tasks, solidifying its advantage as a promising base model for further SFT and DPO. Finally, on the individual tasks where Llama-3.3 does achieve better performance than Llama-3.1, some of the differences are (extremely) small, such as SCOTUS, CaseHOLD and COLIEE Task-4. Based on the results, we conducted all our training by utilizing Llama-3.1-70B-Instruct as the base model.

Benchmark	Task	3.3	3.1
LexGLUE	ECtHR A★	69.2	<u>73.6</u>
	ECtHR B★	76.2	<u>80.0</u>
	EUR-LEX★	31.7	<u>33.9</u>
	LEDGAR★	<u>65.8</u>	63.6
	SCOTUS★	<u>67.7</u>	67.5
	CaseHOLD§	<u>71.6</u>	<u>71.6</u>
	Average	63.7	**65.0**
COLIEE	Task-1★	<u>32.6</u>	30.8
	Task-2★	49.2	<u>57.7</u>
	Task-3△	60.7	<u>61.3</u>
	Task-4§	<u>78.9</u>	78.4
	Average	55.4	**57.0**
δ-Stance†	3-class	54.5	<u>55.1</u>
	4-class	41.3	<u>42.6</u>
	Average	47.9	**48.8**
LegalBench▲		76.7	**78.0**
SuperGPQA (Law)§		39.9	**40.3**

★ Micro-F1
† Macro-F1
△ Macro-F2
§ Accuracy
▲ Balanced Accuracy

Table 7 Comparison between the out-of-the-box version of Llama-3.3-70B-Instruct and Llama-3.1-70B-Instruct, denoted as 3.3 and 3.1 respectively.

E Prompts

E.1 IRAC KG Generation

Figure 5 and Figure 6 contain the prompt for producing the IRAC KG. Due to its length, we split it to two parts. We created the KGs using U.S. legal case opinions. Although these cases may contain personally identifying information, they are public data and are searchable via various online tools.

E.2 Prompt for Generating the Explanation Section of the SFT Data

Figure 7 and Figure 8 show the prompt for producing the SFT training data. Due to its length, we split it to two parts.

E.3 Prompt for the LLM Judge during DPO Data Generation

Figure 9 presents the LLM Judge prompt for producing the DPO training data.

Prompt for Generating IRAC KG (Part I)

You are tasked with building a knowledge graph by extracting entities and relations from a given legal case using the IRAC (Issue, Rule, Application, Conclusion) framework. This knowledge graph will be used for legal reasoning. Follow these instructions carefully to complete the task.

First, carefully read and analyze the following legal case:

<legal_case>
{case_opinion}
</legal_case>

Now, follow these steps to construct the knowledge graph:

1. Entity Extraction:
Extract the following types of entities from the legal case:
- Case: The given case itself
- CitedCase: Other cases cited in the given case
- MaterialFact: Factual elements relevant to the case
- LegalIssue: Legal issues presented in the given case
- Conclusion: The court's ruling on a specific LegalIssue
- Rule: Legal rules or principles applied in the case
- Statute: Laws referenced in the case
- Regulation: Regulations mentioned in the case

For each entity, assign a unique ID, determine its type, and provide a succinct text representing the key aspects of the entity.

2. Relation Extraction:
Identify and extract the following relations between the entities:
- CITES: From Case to CitedCase
- REFERENCES: From Case to Statute or Regulation
- ARISES_FROM: From LegalIssue to MaterialFact
- ADDRESSES: From Rule to LegalIssue
- APPLIED_TO: From Rule to MaterialFact
- DERIVES_FROM: From Rule to CitedCase, Statute, or Regulation
- LEADS_TO: From Rule to Conclusion

For each relation, assign a unique ID, determine its type, and identify the source (from) and target (to) entities.

Figure 5 Prompt for generating IRAC KG (Part I).

Prompt for Generating IRAC KG (Part II)

3. Output Formatting:
Present your output as a JSON-formatted string that adheres to the following pydantic model structure. Ensure that all required fields (id_, type_, label_ for entities; id_, type_, from_, to_ for relations) are properly filled.

```
class Entity(BaseModel):
id_: str = Field(description="entity ID")
type_: str = Field(description="entity type")
label_: str = Field(description="a succinct text representing the key aspects of the entity")

class Relation(BaseModel):
id_: str = Field(description="relation ID")
type_: str = Field(description="relation type")
from_: str = Field(description="source of relation")
to_: str = Field(description="target of relation")

class KG(BaseModel):
vertices_: List[Entity] = Field(description="vertices from KG")
relations_: List[Relation] = Field(description="relations from KG")
```

4. Ensure that your output is comprehensive, capturing all relevant entities and relations from the given legal case. Be thorough in your analysis and precise in your entity and relation extractions.

5. For entity type "CitedCase", only extract the most important cited cases that are relevant for resolving the legal issues.

6. Do not generate any additional text, explanation or analysis.

Figure 6 Prompt for generating IRAC KG (Part II).

Prompt for Generating the SFT Training Data (Part I)

You are tasked with producing Instruction Finetuning (IFT) data for training a Large Language Model (LLM). The goal is to create a dataset that will teach the model how to identify legal issues and applicable rules based on case facts. This task is crucial for developing AI systems capable of assisting in legal analysis.

First, you will be presented with the facts of a specific legal issue in the legal case. Read these carefully:

<case_facts>
{material_facts}
</case_facts>

Analyze these facts thoroughly, considering potential legal issues that might arise from the situation described and potential legal rules that might be applicable to the facts and the legal issues.

Now, here is one actual legal issue identified for this case, and the actual legal rules that are applicable to the given facts and the legal issue.

<legal_issue>
{legal_issue}
</legal_issue>

<rules>
{rules}
</rules>

Your task is to create IFT data that will teach an LLM to derive this issue from the given facts, and also to identify these legal rules that are applicable to the given case facts and the legal issue. To do this, follow these steps:

1. Write a basic instruction for the LLM without including any details.

2. In your response, include the above case facts, the legal issue, and the identified rules exactly as they were provided to you.

3. Instruct the LLM to provide a brief explanation for the legal issue by relating it back to the given facts, and also on why the rules are applicable to the given case facts and the legal issue.

Figure 7 Prompt for generating the SFT training data (Part I).

Prompt for Generating the SFT Training Data (Part II)

4. Format your output using the following XML format, and do not add any newlines or extra spaces between the XML elements:

```
<sft_data>
    <sft_input>
        <instruction>
            [Your instruction here]
        </instruction>
        <case_facts>
            [The list of case facts:
            use separate <fact> tags for each case fact]
        </case_facts>
        <output_format>
            Provide your response in pretty print XML. Use top tag
            legal_analysis, and the following sub-tags:
            legal_issue: [The legal issue here]; rules: [A list of rules];
            explanation: [Brief explanation of the legal issue by relating
            it to the case facts, and also on why the rules are applicable
            to the given case facts and the legal issue].
        </output_format>
    </sft_input>
    <sft_output>
        <legal_issue>
            [The legal issue here]
        </legal_issue>
        <rules>
            [List each applicable rule using the tag rule]
        </rules>
        <explanation>
            [Brief explanation of the legal issue by relating it to the
            case facts, and also on why the rules are applicable
            to the given case facts and the legal issue]
        </explanation>
    </sft_output>
</sft_data>
```

Now, create the IFT data for the case facts, the legal issue, and the rules provided earlier. Ensure your instruction is clear and comprehensive, and that your output follows the above XML format.

Do not generate any additional text.

Figure 8 Prompt for generating the SFT training data (Part II).

LLM Judge Prompt for Generating the DPO Training Data

You are a legal analyst tasked with determining whether specific legal rules can be applied to address a given legal issue.

Your Task:
Analyze whether the provided "Rules to Evaluate" are applicable to the stated legal issue, given the case facts and context.

Information Provided:
Case Facts (the relevant factual circumstances of the case):
{case_facts}

Legal Issue (the specific legal question or dispute that needs to be resolved):
{legal_issue}

Known Applicable Rules (a set of legal rules/statutes/precedents that are already established as relevant to this legal issue):
{chosen_rules}

Rules to Evaluate (another set of legal rules that need to be assessed for applicability):
{rejected_rules}

Your Analysis Should Address:
- Relevance: Do the "Rules to Evaluate" address the same legal subject matter or related issues as the stated legal issue?
- Factual Alignment: Are any factual elements required by these rules present in the given case facts?
- Legal Compatibility: Are these rules consistent with or complementary to the "Known Applicable Rules"?
- Jurisdictional Considerations: Are there any jurisdictional or procedural requirements that affect applicability?
- Scope and Limitations: Are there any known boundaries to how these rules might apply?

Provide your conclusion in JSON format. Use Rules as the top-level attribute; underneath it, include the evaluation for each rule in "Rules to Evaluate" with the following three attributes:
- "Rule": the rule that is being assessed; use the rule text exactly as provided
- "Applicability": Whether each rule in the "Rules to Evaluate" set is applicable (Yes/No/Potentially)
- "Reasoning": Your reasoning for each determination

Figure 9 LLM Judge prompt for generating the DPO training data.